eg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

AMTCS202 – ALLIED MATHEMATICS - II

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Evaluate $\int_0^a \int_0^b (x + y)dxdy$.
2. Find $L(3cos4t)$.
3. Solve $(D^2 - 9)y = 0$.
4. If the vector $\vec{F} = (x + 3y)\hat{\imath} + (y - 2z)\hat{\jmath} + (x + az)\hat{k}$ is solenoid, then find a.
5. What is the relation between Δ & E.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Evaluate $\int\int y^2\,dxdy$ over the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
7. Find $L[te^{-t}sint]$.
8. Solve $(D^2 - 4D + 1)y = x^2 + e^{2x}$.
9. Prove that $\nabla . r^n\vec{r} = (n + 3)r^n$.
10. Find the missing term in the following data

x	0	5	10	15	20	25
y	7	11	14	-	24	32

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) Evaluate $\int_0^1 \int_0^2 \int_0^3 (x + y + z)dxdydz$.

(or)

b) Evaluate $\int \int x^2\, dxdy$ over the area of the circle $x^2 + y^2 = a^2$.

12. a) Find $L^{-1}\left[\frac{4s^2-3s+5}{(s+1)(s-1)(s-2)}\right]$.

(or)

b) Solve by Laplace transform method
d$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 4e^{-t}$, $y(0) = y'(0) = 0$.

13. a) Solve $\frac{d^2y}{dt^2} + y = cosecx$ using method of variation of parameters.

(or)

b) Solve $(D^3 - 5D^2 + 8D - 4)y = e^{2x} + x$.

14. a) If $\vec{F} = x^2y\hat{\imath} + y^2z\hat{\jmath} + z^2x\hat{k}$ then find $curl\ curl\ \vec{F}$.

(or)

b) Verify Green's theorem in the plane for
$\int_C (xy + y^2)dx + x^2 dy$ where C is the closed curve of the region bounded by the lines $y = x$ and the parabola $y = x^2$.

15. a) Use Newton's formula to find y when $x = 142$ given that

x	140	150	160	170	180
y	3.685	4.854	6.302	8.076	10.225

(or)

b) Use Lagrange's formula to find y when $x = 2$ given

x	0	3	5	6	8
y	276	460	414	343	110

****